

Justification for Cambridge Research visual display & control systems purchase

Cambridge Research Systems Limited NIH vendor ID 1281203

Brief overview:

The Briggs LSR lab will have 4 “rigs” to perform visual neurophysiology in behaving NHPs (2 identical rigs), behaving ferrets (1 rig) and anesthetized NHPs and ferrets (1 rig). Each rig consists of many components (e.g. computers, data acquisition systems, visual stimulation systems, optogenetics systems, cameras, etc). Components are manufactured by different vendors but are built to synchronize. This quote is for components for 3 of these rigs that are necessary for visual stimulus display and experimental control. All of these components are manufactured by Cambridge Research Systems. See bottom section for line-item descriptions.

Need for Briggs LSR lab:

Each of the rigs requires its own visual stimulus design/display and experimental control system. Each of these systems includes many parts (computers, displays, cables, software, calibration unit) and most of these components are not portable between rigs (the calibration unit is portable, so only 1 calibration unit is requested). I would like to purchase these high-quality, state-of-the-art systems that will last many years (see sections below for more details).

Justification for Cambridge Research Systems:

There are ~2 competing manufacturers of precision visual stimulus display and experimental control systems that would support my research and Cambridge Research is a leader in the market based on quality and support. I have used Cambridge Research systems since 2011 (many of my systems are still working perfectly after 14 years). I want to continue using Cambridge Research systems because I am familiar with them, they last, and they synchronize with the other rig components. Furthermore, their customer support will ensure a smoother transition by providing assistance programming my new rigs.

Justification for LCD-compatible systems:

It used to be that CRT monitors were the highest standard in visual stimulus displays available in terms of precise timing control and fast, accurate graphics display. Cambridge Research has migrated to LCD-based systems as CRTs are no longer produced. I want to work with Cambridge Research to build my visual stimulus generation and experimental control systems using displays that will be supported in the future.

Line-item descriptions from quote:

1. LCD display monitors (3): used for visual stimulus display to subjects
2. Calibration unit (1): required to calibrate displays for color stimulus display
3. Computer workstations (3): required to program visual stimuli and control experiments
4. Software installation and support (1 each): Cambridge needs to set up each system in-house but provides extensive support including programming assistance
5. Shipping costs